

ODENSE AP, Denmark

WMO: 061200

Lat: 55.475N

Lon: 10.331E

Elev: 17

StdP: 101.12

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.4	(c) -5.3	(d) -9.9	(e) 1.6	(f) -5.2	(g) -8.0	(h) 1.9	(i) -4.0	(j) 15.1	(k) 6.5	(l) 13.7	(m) 6.9	(n) 3.1	(o) 120	(p) 0.650

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.5	(c) 27.1	(d) 19.0	(e) 25.1	(f) 18.3	(g) 23.3	(h) 17.5	(i) 20.6	(j) 24.2	(k) 19.5	(l) 23.0	(m) 18.5	(n) 21.9	(o) 4.8	(p) 120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.3	14.0	22.0	18.2	13.1	20.9	17.2	12.3	19.8	59.2	24.2	55.5	23.0	52.4	22.0	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 12.1	(b) 10.7	(c) 9.6	(e) -11.7	(f) 29.8	(g) 4.5	(h) 2.1	(i) -14.9	(j) 31.3	(k) -17.6	(l) 32.6	(m) -20.1	(n) 33.8	(o) -23.4	(p) 35.3		
(a) 12.1	(b) 10.7	(c) 9.6	(e) -11.9	(f) 22.6	(g) 4.3	(h) 1.3	(i) -15.0	(j) 23.5	(k) -17.6	(l) 24.2	(m) -20.0	(n) 24.9	(o) -23.1	(p) 25.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	1.5	1.9	3.9	7.6	11.6	14.7	17.3	17.2	14.0	9.7	5.7	2.8
	DBStd	6.47	3.69	3.76	3.25	3.16	2.97	2.73	2.63	2.53	2.43	2.94	3.14	3.96
	HDD10.0	1182	263	226	191	86	17	1	0	0	2	41	132	223
	HDD18.3	3443	522	459	449	323	209	113	54	54	132	268	379	482
	CDD10.0	825	0	0	1	13	67	142	225	223	120	31	3	1
	CDD18.3	45	0	0	0	0	1	4	21	18	1	0	0	0
	CDH23.3	396	0	0	0	2	9	54	186	138	6	0	0	0
	CDH26.7	65	0	0	0	0	1	8	32	24	0	0	0	0
Wind	WSAvg	4.8	5.4	5.6	5.4	4.8	4.5	4.5	4.1	4.3	4.6	4.7	4.9	5.2
Precipitation	PrecAvg	652	60	43	38	33	45	63	58	70	62	66	57	58
	PrecMax	818	117	97	89	83	79	146	117	140	142	162	129	124
	PrecMin	451	2	6	8	2	19	2	11	23	19	23	13	17
	PrecStd	99	31	20	20	19	18	32	31	35	34	28	29	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.8	10.8	14.2	21.2	24.4	27.9	29.5	29.0	24.2	18.8	14.1	11.2
		MCWB	9.0	8.8	9.3	14.3	17.4	20.0	19.8	19.4	17.7	15.3	13.5	10.5
	2%	DB	8.4	9.0	11.9	17.7	21.4	24.8	27.2	27.0	21.9	16.2	12.4	10.0
		MCWB	7.8	8.0	8.5	12.0	15.1	18.3	18.9	18.8	16.8	14.2	11.6	9.4
	5%	DB	7.4	8.0	9.9	15.2	19.4	22.4	25.4	24.6	19.9	15.0	11.2	8.9
		MCWB	6.8	7.1	7.7	10.7	14.3	16.9	18.4	18.2	16.0	13.6	10.4	8.3
	10%	DB	6.4	6.9	8.5	13.2	17.7	20.6	23.6	22.7	18.2	14.0	10.1	7.9
		MCWB	5.9	6.1	6.9	9.7	13.2	15.9	17.7	17.6	15.2	12.8	9.4	7.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.2	9.4	10.8	15.2	18.9	22.2	22.1	21.5	19.3	16.2	13.7	10.7
		MCDB	9.5	10.0	12.7	19.9	22.8	26.9	25.6	25.2	22.2	17.9	14.0	11.0
	2%	WB	7.9	8.1	9.1	12.9	16.6	19.5	20.6	20.3	17.7	14.8	11.8	9.6
		MCDB	8.3	8.7	10.8	16.1	19.6	22.9	24.9	23.3	20.0	16.0	12.3	10.0
	5%	WB	6.8	7.1	8.1	11.5	15.2	17.8	19.5	19.2	16.7	13.8	10.6	8.4
		MCDB	7.2	7.7	9.3	14.3	18.1	21.0	23.4	22.6	18.9	14.8	11.1	8.8
	10%	WB	5.9	6.1	7.1	10.2	13.9	16.6	18.5	18.4	15.8	13.0	9.4	7.3
		MCDB	6.3	6.7	8.3	12.5	16.6	19.6	22.1	21.6	17.9	13.9	9.9	7.8
Mean Daily Temperature Range	5% DB	MDBR	4.0	4.4	6.2	8.2	9.0	9.1	9.5	8.9	7.3	5.9	4.4	4.0
		MCDBR	4.3	5.0	8.6	11.6	12.5	12.4	12.8	12.3	9.9	6.8	4.7	4.1
		MCWBR	4.1	4.1	5.8	6.6	7.3	7.0	6.1	5.7	5.7	5.2	4.1	4.0
	5% WB	MCDBR	4.2	4.7	7.3	10.4	10.7	11.1	11.2	10.3	8.7	6.4	4.3	4.0
		MCWBR	4.1	4.1	5.4	6.5	7.2	7.2	6.2	5.9	5.6	5.1	4.0	3.9
Clear-Sky Solar Irradiance	taub	0.318	0.338	0.363	0.398	0.391	0.384	0.403	0.403	0.378	0.357	0.335	0.305	
	taud	2.274	2.303	2.285	2.238	2.329	2.375	2.339	2.351	2.402	2.406	2.327	2.252	
	Ebn at Noon	591	726	797	820	852	861	834	809	778	700	567	511	
	Edh at Noon	56	78	100	120	116	113	115	107	89	71	54	46	
All-Sky Solar Radiation	RadAvg	0.46	1.07	2.49	4.14	5.32	5.70	5.29	4.23	2.99	1.50	0.59	0.33	
	RadStd	0.08	0.16	0.23	0.49	0.51	0.42	0.56	0.29	0.29	0.19	0.09	0.03	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (63 neighbors)	+0.39	N/A	N/A	N/A	N/A	N/A	-116	-169	N/A	N/A

Nomenclature: See separate page